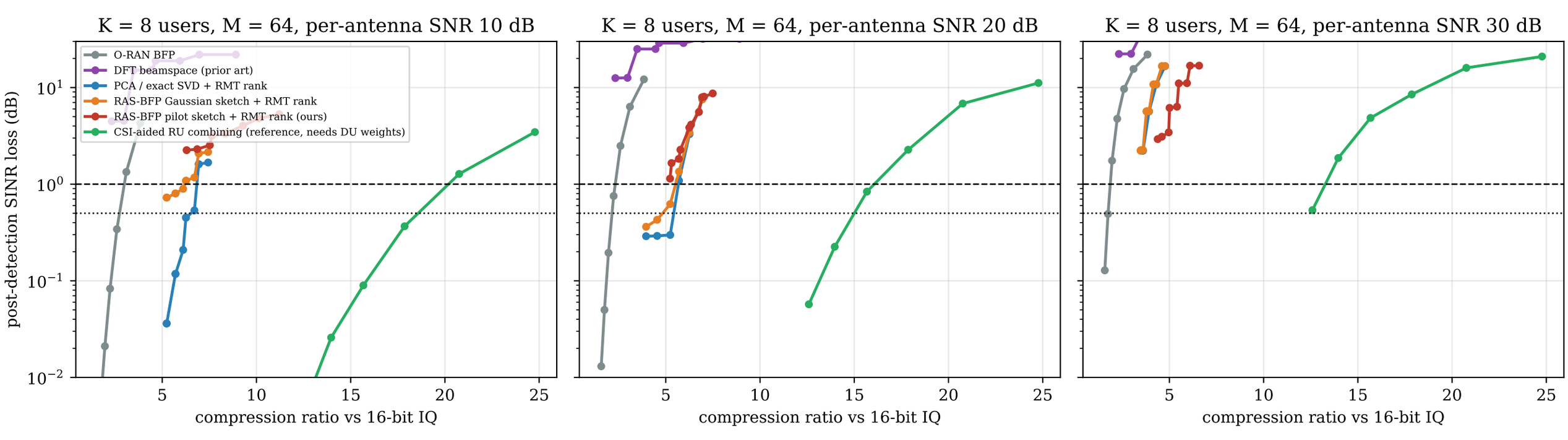
K = 8 users, M = 64, per-antenna SNR 10 dB
K = 8 users, M = 64, per-antenna SNR 20 dB
K = 8 users, M = 64, per-antenna SNR 30 dB
O-RAN BFP
DFT beamspace (prior art)
PCA / exact SVD + RMT rank
RAS-BFP Gaussian sketch + RMT rank
RAS-BFP pilot sketch + RMT rank (ours)
CSI-aided RU combining (reference, needs DU weights)
post-detection SINR loss (dB)
10^{1}
10^{0}
10^{-1}
10^{-2}
5
10
15
20
25
compression ratio vs 16-bit IQ